

1 Reviewer zkCU (Rating: 4)

Thank you very much for your review and constructive comments. Here we would like to address the reviewer's concerns and hope that this helps improve your evaluation of our paper.

Weakness #1: Although the method does not require an explicit reward function, it still relies on a preference annotator and iterative online preference collection. This means that the practical cost is shifted rather than fully removed. In particular, I think the paper should discuss more carefully the sample efficiency and practical convergence behavior of the method. For example: how many preference annotations are needed before the method becomes competitive, how many iterations are typically required in practice, and whether the online refinement process remains feasible when preference labels are expensive.

Question #1: how many preference annotations are needed before the method becomes competitive? The following are the accuracy results when trained using 8 samples per example (4 preference pairs), and 4 samples per example (2 preference pairs).

- (sample count = 8) pass@1: 68.69, maj@8: 82.03
- (sample count = 4) pass@1: 42.10, maj@8: 75.05

Given that the reference model achieves pass@1: 69.1, maj@8: 85.8, 8 samples is a good threshold empirically.

Our Response:

Question #2: how many iterations are typically required in practice?

Our Response: As seen in the accuracy results in Table 1, and in our Ablation study in the appendix Sec E.1 [Table 1], the performance gains over the base model is the highest after the first iteration compared to the second.

Question #3: whether the online refinement process remains feasible when preference labels are expensive.

Our Response: We believe our algorithm can be adapted to an off-policy algorithm where the preference function virtue of the guidance model can be learnt offline. That is, as long as one has access to preference data, the training process of multiple iterations can be collapsed into one iteration. However, a fundamental assumption made in such offline algorithms is sufficient coverage of the offline dataset, and would warrant a separate analysis.

2 Reviewer EMXM (Rating: 3)

Thank you very much for your review and constructive comments, as well as for giving a positive rating to our work. We address your concerns below and hope our clarifications can further strengthen your impression of our submission.

Weakness #1:

Theoretical assumptions are restrictive and poorly connected to practice. The regret bound in Theorem 2 depends on a linear reward assumption (Assumption 1) and a boundedness condition on the expected embedding under any learned policy (Assumption 2). Neither assumption is verified nor even discussed in the context of the actual LLM experiments. Specifically, Assumption 2 implicitly rules out deterministic policies and requires some exploration — it is unclear how this applies to Llama-3 8B in practice. The gap between theory and experiment needs to be addressed more clearly.

Our Response:

We thank the reviewer for highlighting this point. In our implementation, the preference function is parametrized as a single linear head on top of the LLM backbone, which directly matches the linear reward assumption in our theory.

Regarding the boundedness assumption, the embeddings correspond to final-layer representations of the LLM, are empirically well behaved and bounded in practice due to standard normalization and training dynamics. Below we present the embedding, and gradient norm (guidance model only) statistics observed throughout our training and evaluation pipeline

49 Training Phase

50 • Hidden representation Statistics

51 – Reference Model: mean 155.15, std 5.03, min/max 93.34/162.77

52 – Guidance Model: mean 48.10, std 8.74, min/max 34.70/107.52

53 • Gradient Norm Statistics

54 – Guidance Model: mean 11.19, std 8.42, min/max 0.69/87.5

55 Evaluation Phase

56 • Hidden representation Statistics

57 – Reference Model: mean 155.13, std 5.06, min/max 90.39/162.56

58 – Guidance Model: mean 46.11, std 5.49, min/max 38.36/88.77

59 The above statistics empirically validate the boundedness of the hidden embedding. For the guidance
60 model, we observe a much wider spread than the reference but no runaway growth compared to the
61 reference model. This is expected since we are training the guidance model to learn the preference
62 function but not the reference model. The differences in norm scales can be attributed to differences
63 in size, depth, hidden width etc. between the 7B(reference model) vs 1B(guidance model).

64 Norms remain bounded in practice

65 The boundedness of final-layer hidden representations in LLM training follows from a combination
66 of normalization layers, residual architecture, and optimization dynamics.

- 67 • Most LLMs use RMSNorm (applicable for LLaMa family) or LayerNorm that controls
68 magnitude of representation.

$$\text{RMSNorm}(h) = \frac{h}{\sqrt{\frac{1}{d} \sum_i h_i^2 + \epsilon}} \cdot \gamma$$

69 where γ is a learnable parameter that remains stable in practice.

- 70 • Residual connections preserve boundedness. In standard transformer architecture,

$$h_{l+1} = h_l + f(\text{Norm}(h_l)) \quad (1)$$

71 where l is the layer index. The inputs to $f(\cdot)$ are normalized (via RMSNorm/LayerNorm),
72 and $f(\cdot)$ remains Lipschitz under standard parameter scales, resulting in bounded updates.

- 73 • Adaptive Optimizers like Adam (used in our training pipeline) enforce gradient norm stability
74 which actively discourages large activation norms.

75 Additionally, we provide sensitivity analysis with respect to model capacity in response to Question 4
76 below.

77 Weakness #2:

78 The preference oracle is a stronger external model — this tension is underexplored.

79 In the GSM8K experiments, preference labels are provided by OpenMathInstruct-2, a model specifi-
80 cally tuned for mathematical reasoning and arguably more capable than the reference policy on this
81 task. This partly reintroduces a dependency on a strong external model, which somewhat undermines
82 the paper’s framing of "eliminating the reward model." A natural question arises: what happens when
83 the preference oracle is weaker than or comparable to the reference model? This should be discussed.

84 Our Response: [Update for response to rebuttal]:

85 We would like to point out that the 70B Llama model is not used as a preference oracle during
86 training; rather it is a tool for evaluation of the trained model (serving as a judge of response quality).
87 Below we explain the setup for the AlpacaEval experiment in greater detail:

88 For training Q# and PITA, we take offline human-labeled data from AlpacaFarm [1] where each
89 sample consists of an instruction, a pair of responses, and a preference label. The preference labels

come from multiple human annotators recruited from Amazon Mechanical Turk, so we can expect noise in the labels due to differing opinions between the annotators. To provide a reward signal for Q#, we take the preference label and convert it to a binary reward for each response (1 if preferred, 0 otherwise).

The AlpacaEval [2] framework enables us to use LLM-as-a-judge for evaluation when human evaluators are not available to judge responses from the trained models. We use the weighted_alpaca_eval_vllm_llama3_70b annotator to compare responses generated by the candidate algorithms to responses generated by the text_davinci_003 model in a reproducible manner. Specifically, a locally-hosted Meta-Llama-3-70B-Instruct model is shown the query and two responses, and it is directed to output one of two different tokens depending on which response would be preferred by humans. The log probabilities which the model assigns to those tokens is used to determine the final binary preference. A win rate is determined based on the binary preferences over 805 queries, and a length-controlled win rate is calculated according to the method in [2]. This is the basis for the scores reported in our initial rebuttal.

[1] Yann et. al. (2023). AlpacaFarm: A Simulation Framework for Methods that Learn from Human Feedback. [2] Yann et. al. (2024). Length-Controlled AlpacaEval: A Simple Way to Debias Automatic Evaluators. arXiv preprint arXiv:2404.04475.

Algorithm	length-controlled win rate (%)	win rate (%)
π_{ref}	46.91	45.1
Q#	47.15	45.95
PITA	47.22	46.14

Weakness #3:

No variance or statistical significance reported.

All tables report only point estimates across all tasks and baselines. Given that PITA uses temperature sampling ($T=0.8$, $p=0.9$) at inference time and involves stochastic rollouts during training, both sampling variance and training variance can be non-trivial. The gap between PITA (77.11) and Q# (78.4) on GSM8K pass@1, for instance, could be statistically insignificant without error bars. At a minimum, results across 3 random seeds should be reported, with standard deviations.

Our Response: We present the results across 3 random seeds in our response to Question 1. We would be glad to add the error bars in our revised submission.

Weakness #4:

Sample efficiency claim is not cleanly supported.

The evidence for sample efficiency is still somewhat indirect. A large part of the benefit comes from the brittleness of Q#-HF, rather than from a clean controlled comparison showing that PITA consistently dominates across matched data budgets.

Our Response: We would be glad to add a comparison with the baseline Q#-HF in our revised submission. Additionally, we present our ablation results for PITA as number of samples per example are varied in our response to Question 4 below.

Minor Algorithm 1, Line 8: Completions are sampled from rather than the current policy. A brief discussion of this choice and its effect on convergence would be helpful.

Our Response: This is a direct consequence of the optimal Q, and V functions presented in Equation 11, Theorem 1. Since the expectation is with respect to completions from the reference model, we retain the same in our implementation.

Minor: Algorithm 1, Line 8: Completions are sampled from the reference model rather than the current policy. A brief discussion of this choice and its effect on convergence would be helpful.

Our Response: This follows directly from the optimal Q and V formulations (Eq. 11, Theorem 1), where expectations are taken over completions from the reference model. Accordingly, we sample from the reference policy in practice. This choice ensures consistency with the theoretical objective.

Question #1:

136 All tables report point estimates only, with no standard deviations across random seeds. Could the
137 authors report means and standard deviations across at least 3 random seeds?

138 **Our Response:**

139 Here we present the results for our algorithm across 8 different seeds. PITA:

- 140 • pass@1: 0.7662 ± 0.0099
- 141 • maj@8: 0.8572 ± 0.0031

142 We will be glad to provide the mean/std results across multiple seeds for all the baselines, and
143 algorithms in the revised manuscript.

144 **Question #2:**

145 Can the authors provide a cleaner end-to-end efficiency comparison, including both training and
146 inference cost, against baselines that also rely on small auxiliary models rather than only against full
147 post-training methods?

148 **Our Response:**

149 Here we present the end-to-end efficiency comparison including both training and inference costs
150 against baselines that also rely on small auxiliary models like Q# [Zhou et al., 2025], and CD [Mudgal
151 et al., 2024].

152 Each of the algorithms are run on a single H100 GPU, for one epoch of training, and eval on train/test
153 suites of the GSM8k dataset. A step-level record corresponds to one logged optimizer-update event
154 (i.e., one update after gradient accumulation), where we measure wall-time and throughput for that
155 specific update interval. An example-level metric is computed per inference request (prompt + guided
156 generation). Reference model, Guidance model, and number of optimizer steps are kept consistent
157 across all the algorithms.

Metric	PITA	QSharp	CD
Average Step time (s)	0.1519	0.1581 (+4.1%)	0.1525 (+0.4%)
Step throughput (examples/s)	56.12	53.80 (-4.1%)	55.82 (-1.0%)
Step throughput (tokens/s)	12401	11,895 (-4.1%)	12,339 (+0.0%)
Per-example training latency (mean, s)	0.01898	0.01976 (+4.1%)	0.01906 (+0.4%)
Run-level throughput (examples/s)	47.48	44.41 (-6.5%)	45.74 (-3.7%)
Run-level throughput (tokens/s)	10,609	9921 (-6.4%)	10,218.65 (-3.7%)
Overall training cost (GPU hours)	0.472	0.505 (+6.9%)	0.490 (+3.8%)
Overall training cost (Total wall-clock time) (s)	1699.94	1817.8 (+6.9%)	1764.89 (+3.8%)

Table 1: Comparison of QSharp and CD against PITA (same dataset and model scale).

Metric	PITA	QSharp	CD
Overall inference cost (total wall-clock time, s)	10,290	9,979.10	10,452
Overall inference cost (GPU hours)	2.86	2.77	2.90
Total generated tokens	1,559,842	1,511,414	1,517,496
End-to-end latency per example (mean $\pm$ std, s)	0.9753 ± 0.4050	0.9457 ± 0.3889	0.9905 ± 0.4644
Generation-only latency per example (mean $\pm$ std, s)	0.9022 ± 0.3649	0.8758 ± 0.3535	0.9201 ± 0.4218
Generated tokens per example (mean $\pm$ std)	147.82 ± 67.37	143.23 ± 65.96	143.81 ± 70.06
Throughput (run-level, tokens/s)	151.58	151.46	145.18
Guidance calls per example (mean $\pm$ std)	31.60 ± 11.85	30.61 ± 11.45	31.37 ± 13.34
Reference-only compute estimate (FLOPs)	2.505×10^{16}	2.427×10^{16}	2.437×10^{16}
Guided compute estimate (FLOPs)	1.525×10^{17}	1.438×10^{17}	1.484×10^{17}

Table 2: Comparison of evaluation-phase efficiency metrics for PITA, QSharp, and CD.

158 **Question #3:** The arithmetic experiments use model-based preference feedback. How robust is the
159 method when the preference data is genuinely noisy, inconsistent, or multi-annotator human feedback
160 rather than model-generated labels?

	Algorithm	length-controlled win rate (%)	win rate (%)
161 Our Response:	π_{ref}	46.91	45.1
	Q#	47.15	45.95
	PITA	47.22	46.14

162 We would like to point out that the 70B Llama model is not used as a preference oracle during
163 training; rather it is a tool for evaluation of the trained model (serving as a judge of response quality).
164 Below we explain the setup for the AlpacaEval experiment in greater detail:

165 For training Q# and PITA, we take offline human-labeled data from AlpacaFarm [1] where each
166 sample consists of an instruction, a pair of responses, and a preference label. The preference labels
167 come from multiple human annotators recruited from Amazon Mechanical Turk, so we can expect
168 noise in the labels due to differing opinions between the annotators. For Q#, we take the preference
169 label and convert it to a binary reward for each response (1 if preferred, 0 otherwise).

170 The AlpacaEval [2] framework enables us to use LLM-as-a-judge for evaluation when hu-
171 man evaluators are not available to judge responses from the trained models. We use the
172 weighted_alpaca_eval_vllm_llama3_70b annotator to compare responses generated by the candidate
173 algorithms to responses generated by the text_davinci_003 model (which is the default reference in
174 AlpacaEval). Specifically, a locally-hosted Meta-Llama-3-70B-Instruct model is shown the query and
175 two responses, and it is directed to output one of two different tokens depending on which response
176 would be preferred by humans. The log probabilities which the model assigns to those tokens is used
177 to determine the final binary preference. A win rate is determined based on the binary preferences
178 over 805 queries, and a length-controlled win rate is calculated according to the method in [2]. This
179 is the basis for the scores reported in our initial rebuttal.

180 [1] Yann et. al. (2023). AlpacaFarm: A Simulation Framework for Methods that Learn from
181 Human Feedback. [2] Yann et. al. (2024). Length-Controlled AlpacaEval: A Simple Way to Debias
182 Automatic Evaluators. arXiv preprint arXiv:2404.04475.

183 **Question #4:**

184 How sensitive is PITA to the number of refinement rounds , the number of sampled completions per
185 context, and the capacity of the preference model backbone? The ablation in Table 5 only varies the
186 number of rounds; a more systematic sensitivity analysis would help practitioners deploy the method.
187 **Our Response:** Below we present our results for different layers of model backbone while retaining
188 the data scale as used in our original experiments with a single linear head. While the optimal η
189 seems to have shifted, the general trend where the accuracy increases, and falls after the optimal η
190 remains true. The range of accuracy values remain fairly close to the single linear head case.

	η	pass 1 (n = 5)	maj8 (n=5)	pass1 (n=3)	maj8 (n=3)
	0.5	0.6629	0.8059	0.6675	0.7945
	1	0.7023	0.8165	0.7043	0.8264
191	2	0.7536	0.8635	0.7476	0.8529
	4	0.7601	0.8681	0.7472	0.8582
	6	0.7472	0.8673	0.6597	0.8476

192 In the context of number of sampled completions per problem, we see the following accuracy results
193 when trained using 8 samples per example (4 preference pairs), and 4 samples per example (2
194 preference pairs).

- 195 • (sample count = 16) pass@1: 77.11, maj@8: 86.2
- 196 • (sample count = 8) pass@1: 68.69, maj@8: 82.03
- 197 • (sample count = 4) pass@1: 42.10, maj@8: 75.05

198 Given that the reference model achieves pass@1: 69.1, maj@8: 85.8, 8 samples can be regarded as a
199 competitive threshold for the algorithm.

200 **3 Reviewer DRgw (Rating: 3)**

201 We sincerely appreciate the time you took to engage with our work, and we respond to each of your
202 comments below.

Weakness #1: Compared with Q#, the core optimization objective and modeling assumptions of PITA remain largely consistent with prior work. Therefore, its contribution is more incremental than foundational.

Our Response: While it's true that the optimization objective we consider is consistent with prior work, and hence the nature of the optimal policy we achieve. Our work generalizes to preference modeling beyond the standard reward models used to score completions of an LLM, requiring a separate non-trivial analytical framework compared to the binary reward case considered in Q#.

Weakness #2: The empirical evaluation relies primarily on Llama 3 family models (1B–8B) and smaller baselines (GPT-2, TinyLlama). To better assess the generality of PITA, it would be valuable to include experiments with larger-scale models (e.g., 70B) or alternative architectures (e.g., Qwen, Mistral), especially given that inference-time guidance methods may exhibit different scaling behaviors across model families and sizes.

Our Response: Work is motivated by the desire to run small models on edge devices. Hence our desire is to focus on small less performant models which can be modified using our tilting approaches. We use Lama is a representative model class which has reasonably good performance across a variety of tasks.

Weakness #3: The current evaluation mainly focuses on tasks with binary correctness signals, which do not fully reflect the claimed advantage of PITA in settings where scalar rewards are difficult to define. I recommend including additional experiments on domains characterized by ambiguous or subjective preferences, such as open-ended dialogue or helpfulness benchmarks (e.g., AlpacaEval), to better demonstrate the practical benefits of avoiding explicit reward modeling.

	Algorithm	length-controlled win rate (%)	win rate (%)
Our Response:	π_{ref}	46.91	45.1
	Q#	47.15	45.95
	PITA	47.22	46.14

Weakness #4: Empirical results show that KL divergence can be higher compared to Q# algorithm, which is not theoretically analyzed. A deeper investigation of KL dynamics would strengthen the technical contribution (e.g., providing theoretical analysis explaining this behavior or conducting additional experiments across different model scales or architectures to assess its robustness).

Our Response: We attribute the higher KL divergence(over Q#) seen, as a fundamental characteristic of preference modeling in general. It is common to model the preference data using the Bradley Terry model assumption. This naturally results in a unique preference distribution for a family of reward models. We believe the higher KL divergence is a consequence of optimizing for a class of reward models in contrast to a specific reward model like in Q#

Weakness #5: The manuscript contains minor writing inconsistencies, such as the non-standard reference "appendix section E." in Section 6.

Our Response: We thank the reviewer for pointing out the inconsistencies. We will make sure to address this in our revised submission.

Question #1: PITA shows strong gains on in-distribution GSM8K but limited improvement on out-of-distribution MATH-500. What is the intended purpose of this ID/OOD comparison, and what factors might explain the algorithm's limited generalization under distribution shift?

Our Response: The comparison is made to test the robustness of our algorithm for OOD tasks when trained using preference information. The MATH500 dataset consists of problems of higher difficulty over GSM8K. Given the nature of MATH500, we believe the limited generalization performance is fair given the accuracy is consistent across all algorithms.

Question #2: The main results report only two training iterations. Could the authors provide more details on PITA's convergence behavior—e.g., performance trajectories or stability analysis across additional iterations—to better connect the theoretical regret bounds with empirical practice?

Our Response:

Question #3: The performance gap between oracle Q# and Q#-HF is attributed to reward estimation instability. Did the authors evaluate the intermediate reward model on established reward-modeling benchmarks?

252 **Our Response:**

253 **4 Reviewer 9Cac (Rating: 4)**

254 Thank you again for your continued engagement and thoughtful follow-up questions. We address
255 each of them below.

256 I have two main concerns regarding the soundness of the paper.

257 **Weakness #1:** First, the paper is focused on preferences, yet the algorithm and experimental setup
258 do not fully align with this theme. To start, the algorithm requires a large number of preferences
259 acquired ONLINE: for each example, at each step after updating the learned preference distribution,
260 and for every sampled continuation. This is heavy. In practice, the method appears impractical for
261 tasks that rely on real human preferences (which is the primary motivation for preference learning).
262 Even if we relax this requirement and use a prompted LLM as an automatic grader, the process
263 remains very heavy. Indeed, the authors focus their experiments on tasks where true preferences
264 are not required: math problems (with an easy binary reward; resolving "ties" with another reward
265 model is not relevant and does not make it a preference task), the graph problem (again involving a
266 verifiable reward), and the sentiment analysis task (which optimizes against a simple BERT-based
267 sentiment model that outputs a sentiment score). In my view, the latter is especially naive and a weak
268 example, and I would consider discarding it.

269 While I still believe the paper should be accepted (but I leave the decision to the AC, which should
270 consider this weakness and if it finds it major, then reject), raising my score would require either a
271 complete reframing of the paper or the inclusion of at least one task where preference is genuinely
272 central (e.g., using an LLM-as-a-judge to simulate preferences and validating alignment against a
273 small human-labeled subset).

274 **Our Response:**

275

276 **Weakness #2:** Second, there is no comparison with inference-time scaling methods. As I understand
277 it (please correct me if I am wrong, as the generation procedure is not clearly explained), your
278 method involves sampling multiple continuations for each next-token candidate and re-scoring them
279 using the learned preference distribution. If so, it would be important to compare against standard
280 inference-time scaling approaches such as best-of-N and related methods discussed in your appendix.
281 Again, I find the paper strong and the method interesting, but without such comparisons, it is difficult
282 to assess whether the approach is practical or relevant in real-world settings

283 **Our Response:** Our generation procedure involves reweighting the logits of each next-token of the
284 reference model by the smaller guidance model. This follows directly from the structure of the
285 optimal policy considered in Equation 12 where an action a is the next token, and the state s is the
286 context so far.

287 In the context of other inference-time scaling approaches, we view best-of-N and related methods
288 as complementary to our approach. Accordingly, we report performance using maj@8, which is
289 equivalent to best-of-(N) with (N=8). The resulting gains over the pass@1 metric highlight the
290 effectiveness of our strategy.